KELMANS' 1984 PROBLEM ON 3-VERTEX PATH PACKINGS IN CUBIC 3-CONNECTED GRAPHS: EXHAUSTIVE VERIFICATION THROUGH 22 VERTICES

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Abstract. A Λ-packing of a graph is a subgraph all of whose components are 3-vertex paths, and a Λ-factor is a spanning one; $\lambda(G)$ denotes the maximum number of vertex-disjoint 3-vertex paths in G, so that $\lambda(G) \le \lfloor v(G)/3 \rfloor$ always. Kelmans asked in 1984 whether equality holds for every cubic 3-connected graph; specialized to $v(G) \equiv 0 \bmod 6$ this is the Akiyama–Kano conjecture that such a graph has a P_3-factor, and a positive answer would give Reed's domination conjecture for cubic 3-connected graphs. The problem is open, and no computational verification of it at any order appears in the record.

We supply one. For every 3-connected cubic graph on at most 22 vertices — all 6,339,157 of them — $\lambda(G) = \lfloor v(G)/3 \rfloor$, and the applicable strong forms of Kelmans' equivalence theorem hold as well: $(z2), (z3), (z7), (z8)$ at orders $6, 12, 18$; $(t2)$ at $8, 14, 20$; $(f1), (f2)$ at $4, 10, 16$ and $(f1)$ at 22. Those equivalences are constructive, so a single failure at any of these orders would have yielded an explicit cubic 3-connected graph of order divisible by 6 with no Λ-factor. None was found. The graphs were generated with `geng`, and the generated counts agree exactly, at every order, with the published enumerations of connected cubic graphs and of 3-connected cubic graphs. Two pipelines sharing no code — different connectivity tests, different search orders, both negative-controlled — agree on every count and report zero failures, and all 43,580 Λ-factor certificates in the corpus were re-verified from the `graph6` strings alone by standard-library checkers sharing no code with the searcher that produced them (34,429 of them by two such checkers independently). At 24 vertices the search pipeline has completed all 98,101,019 3-connected cubic graphs with no failure and with counts again matching the published enumeration, but the independent recount is outstanding; that order is reported at that strength and no higher. We close with the questions the computation opens.

1. Introduction

1.1. **The question.** All graphs here are finite, undirected, and simple. Write $v(G)$ for the number of vertices of G and Λ for the 3-vertex path P_3. Following [Kel11], a Λ-*packing* of G is a subgraph

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Computation and authorship. The searchers, the independent verifiers, and the audits in this work were produced by **Claude** (Anthropic), directed by the author, on a single Apple M4 laptop. The certificate checkers import only the Python standard library and share no code with the searchers; the adversarial referee of §3.4 was a separate agent instance given no access to the search pipeline's code and wrote its own code throughout. This is a factual methods statement, and it is part of the point of the note: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result.* External tools used by the pipelines: `geng` from `nauty` 2.9.3 [McKP14], Apple clang 17, and Python 3.

Prior-art record. The primary source [Kel11] was read in full (arXiv v2; a copy is archived in the author's working tree, not redistributed here, and all statement numbers below follow its numbering). A novelty sweep was run on August 5 and again on August 11, 2026, the latter immediately before this draft was written: the arXiv API (abstracts mentioning P_3-factors or 3-vertex paths; *Kelmans* in math.CO; *Akiyama–Kano*), the citing literature of [Kel11] through OpenAlex and Semantic Scholar, the Open Problem Garden entry [OPG] fetched live, West's REGS page [West] fetched live, and general web sweeps. The problem is recorded as open in both problem collections, and no computational verification of it at any order was found anywhere in the record. *Draft of August 11, 2026.*

all of whose components are isomorphic to Λ, and a Λ-*factor* is a spanning Λ-packing — what is elsewhere called a P_3-*factor*. Let $\lambda(G)$ be the maximum number of pairwise disjoint 3-vertex paths in G. Counting vertices gives

$$\lambda(G) \;\leq\; \lfloor v(G)/3 \rfloor \tag{1}$$

for every graph, and the question is when (1) is tight.

Problem (A. Kelmans, 1984; Problem 1.10 of [Kel11]). *Is the following claim true?*

(P) *If G is a 3-connected cubic graph, then $\lambda(G) = \lfloor v(G)/3 \rfloor$.*

A cubic graph has even order, so (P) splits into three cases by the residue of $v(G)$ modulo 6: for $v(G) \equiv 0$ the claim is that G has a Λ-factor; for $v(G) \equiv 4$ that some $G - x$ has one; for $v(G) \equiv 2$ that some $G - \{x, y\}$ has one. The first of these is the conjecture of Akiyama and Kano, which we take in the form recorded by West [West] — "when 3 divides n, every 3-connected 3-regular n-vertex graph has a P_3-factor", there attributed to the survey [AK85] — and it is the form in which the problem is posed at the Open Problem Garden [OPG]: does every 3-connected cubic graph on $3k$ vertices admit a partition into k paths on 3 vertices?

Three facts fix the difficulty. First, the Λ-packing problem is NP-hard in general [HK86] and remains NP-hard already for cubic bipartite planar graphs [KMZ05], so (P) is not a statement about an easy optimization problem but about a class on which the optimum is conjecturally forced. Second, 3-connectivity is not a convenience: Kelmans [Kel07] constructs infinitely many 2-connected cubic bipartite planar graphs G with $\lambda(G) < \lfloor v(G)/3 \rfloor$, so the hypothesis cannot be weakened by one unit of connectivity. Third, the problem has a consequence outside path packing. Reed [Reed96] conjectured that a connected cubic graph G has domination number $\gamma(G) \leq \lceil v(G)/3 \rceil$; that conjecture is false for connected and even for 2-connected cubic graphs [Kel06, KS05], but as Kelmans observes [Kel11, §1], if (P) holds then — by way of the strong forms of Theorem 1.4, not directly — Reed's conjecture holds for cubic 3-connected graphs. A *positive* resolution of (P) therefore settles a domination question as a corollary; a counterexample would leave Reed's conjecture on this class untouched.

The problem has been open since 1984. Partial results are known. For every cubic graph $\lambda(G) \geq \lceil v(G)/4 \rceil$ [KM04], and for connected cubic graphs this has been improved to $\lambda(G) \geq \frac{39}{152} v(G)$ when $v(G) \geq 17$ [KMZ08] and to $\lambda(G) \geq \frac{3}{11} v(G)$ when $v(G) \geq 9$ [KZ08], as recorded in [Kel11, 1.3, 1.6, 1.8]. In restricted classes the exact statement is known: $\lambda(G) = \lfloor v(G)/3 \rfloor$ for 2-connected claw-free graphs [KKN01], and claims ($z1$)–($z5$) of Theorem 1.4 below hold for cubic 3-connected claw-free graphs [Kel07b] (see also [Kel11b]). But the general case is untouched, and, as far as our sweeps can determine, no exhaustive verification of (P) at any order has ever been published.

1.2. Result.

Theorem 1.1. *Let G be a 3-connected cubic graph with $v(G) \leq 22$. Then $\lambda(G) = \lfloor v(G)/3 \rfloor$. In particular, every 3-connected cubic graph on 6, 12, or 18 vertices has a Λ-factor, i.e. admits a partition of its vertex set into $v(G)/3$ paths on 3 vertices.*

The statement ranges over all 6,339,157 3-connected cubic graphs of order at most 22; the counts by order are in Table 1. Theorem 1.1 is the base claim (P). The verification also establishes the applicable strong forms of Kelmans' equivalence theorem, which is where its counterexample-hunting power lies (§1.3).

Theorem 1.2. *Let G be a 3-connected cubic graph. Then, in the ranges indicated,*

($z2$) *for $v(G) \in \{6, 12, 18\}$: $G - e$ has a Λ-factor for every $e \in E(G)$;*
($z3$) *for $v(G) \in \{6, 12, 18\}$: for every $e \in E(G)$ some Λ-factor of G contains e;*
($z7$) *for $v(G) \in \{6, 12, 18\}$: $G - X$ has a Λ-factor for every $X \subseteq E(G)$ with $|X| = 2$;*

($z8$) *for* $v(G) \in \{6, 12, 18\}$*:* $G - L$ *has a* Λ*-factor for every* 3*-vertex path* L *in* G*;*
($t2$) *for* $v(G) \in \{8, 14, 20\}$*:* $G - \{x, y\}$ *has a* Λ*-factor for every* $xy \in E(G)$*;*
($f1$) *for* $v(G) \in \{4, 10, 16, 22\}$*:* $G - x$ *has a* Λ*-factor for every* $x \in V(G)$*;*
($f2$) *for* $v(G) \in \{4, 10, 16\}$*:* $G - \{x, e\}$ *has a* Λ*-factor for every* $x \in V(G)$ *and every* $e \in E(G)$ *not incident with* x*.*

Kelmans states ($f2$) without an incidence restriction; restricting it here to edges not incident with x loses nothing, and for the reason his own proof gives: if e is incident with x then $G - \{x, e\} = G - x$, so the incident case is ($f1$) [Kel11, 3.21].

The evidence for both theorems is a complete search: every 3-connected cubic graph of each order was generated, and for each the relevant existence questions were decided exactly — not estimated, not sampled — by an exhaustive procedure (§2.2). The result was then recounted end to end by a second pipeline written independently of the first, and the whole was confirmed by an adversarial referee (§3).

One further order was searched but is *not* part of Theorems 1.1 and 1.2, and is recorded separately at the strength it has reached.

Proposition 1.3. *The search pipeline reports* $\lambda(G) = v(G)/3$ *— that is, a* Λ*-factor — for every one of the* 98,101,019 3*-connected cubic graphs on* 24 *vertices, with no failure on any graph. The enumeration underlying that sweep was audited independently (job accounting reconstructed from the run records, generated counts matched against the published enumerations, zero failure lines in any output), but the graphs themselves have not been re-solved by the independent pipeline. This order is asserted at exactly that strength.*

1.3. **Why the strong forms are the counterexample hunt.** Theorem 1.2 is not a collection of sanity checks. Its statements are the ones Kelmans proves equivalent to (P) as universal claims over the class of cubic 3-connected graphs.

Theorem 1.4 (Kelmans [Kel11, 3.1])**.** *Read as universal statements over the class of cubic* 3*-connected graphs, the nineteen claims* ($z1$)*–*($z9$)*,* ($t1$)*–*($t4$)*,* ($f1$)*–*($f6$) *— conditioned on* $v(G) \equiv 0$*,* 2*,* 4 mod 6 *respectively, and including the seven forms listed in Theorem 1.2 — are pairwise equivalent.*

Two remarks on the statement. Claim (P) is not one of the nineteen, but is equivalent to them, as Kelmans records [Kel11, §1]: ($z1$) is exactly the $v(G) \equiv 0$ case of (P), so (P) implies ($z1$), while conversely ($z1$) yields ($t1$) and ($f1$) by the theorem, and those are stronger than the $v(G) \equiv 2$ and $v(G) \equiv 4$ cases of (P). And the quantification is over the class: the equivalences hold between claims each universally quantified over all cubic 3-connected graphs, not graph by graph. A single graph failing one strong form is therefore not by itself a graph failing another; what it yields is a construction, and that is the next point.

The equivalences are proved by explicit graph compositions — the operations $Aa\sigma bB$ and $Y(\cdot)$ and the graph-compositions $B\{\cdot\}$ of [Kel11, §2], all of which preserve cubicity and 3-connectivity [Kel11, 2.1–2.3] — and Kelmans draws the consequence himself: he gives several different proofs of 3.1, so that "if there is a counterexample C to one of the above claims, then these different proofs provide different constructions of counterexamples to the other claims in 3.1" [Kel11, §1]. The implications therefore run constructively in the direction that matters here: a single cubic 3-connected graph violating any one of ($z2$), ($z3$), ($z7$), ($z8$), ($t2$), ($f1$), or ($f2$) can be converted, by those compositions, into an explicit cubic 3-connected graph of order $\equiv 0$ mod 6 with no Λ-factor, i.e. into a counterexample to (P) and to the Akiyama–Kano conjecture.

This changes what the orders $v(G) \equiv 2, 4$ mod 6 are worth. They are not orders at which the factor question is vacuous; they are orders at which a cheap local test — delete an edge's two endpoints, or one vertex, and ask for a factor of the remainder — is a valid search for a counterexample to the factor case. Of the 6,339,157 graphs covered by Theorem 1.1, only 30,527 have order divisible by 3; the strong forms are what makes the other 99.5% of the corpus evidence rather than context.

Kelmans also records where the strong forms stop [Kel11, (r1)–(r8)], and the boundary is close. Claim ($z7$) is false for $|X| = 3$: a 3-edge cut whose sides have orders not divisible by 3 obstructs any Λ-factor of $G - X$. Claim ($z8$) is tight for two disjoint paths — [Kel11, 6.1] constructs cubic graphs R_s with $v(R_s) = 12s$, cyclically 6-connected for $s \geq 2$ (the smallest member has $c(R_1) = 5$, [Kel11, 6.1(a2)]), having disjoint 3-vertex paths L, L' for which $R_s - (L \cup L')$ has no Λ-factor, while $R_s - L - e$ has one for every edge e and every 3-vertex path L of $R_s - e$. Since $v(R_1) = 12$, the exact boundary of the phenomenon lies inside the verified range: the sweep confirms ($z8$) for every single 3-vertex path at order 12, and Kelmans' construction exhibits the two-path failure at the same order. A verification that reported success on a strong form known to be false would be diagnostic of a broken pipeline; none of the forms tested here is known to be false in the tested range, and none failed.

1.4. **What is not claimed.** We do not claim to resolve Kelmans' problem, the Akiyama–Kano conjecture, or Reed's conjecture for cubic 3-connected graphs. Each is a statement about an infinite class and is untouched by any finite computation; what a proof would have to supply is discussed in §6. We do not claim a bound on the order of a putative counterexample beyond the one Theorem 1.1 gives, namely that none has at most 22 vertices. We do not claim the strong forms outside the orders listed in Theorem 1.2: in particular ($f2$) was not tested at order 22, and ($z7$) and ($z8$) were not tested at order 24. We do not claim order 24 at the strength of Theorem 1.1: Proposition 1.3 is a report of a completed search whose independent recount has not been performed, and it is stated in those terms deliberately. Finally, we claim no priority over the mathematics: the problem, the equivalence theorem, the tightness constructions, and the link to Reed's conjecture are all Kelmans' [Kel11], and the conjecture in its P_3-factor form is Akiyama and Kano's [AK85].

2. The verification

2.1. **Enumeration.** Each order was swept over the output of `geng` from `nauty` 2.9.3 [McKP14], invoked as `geng -q -c -d3 -D3` n, which emits one `graph6` string per isomorphism class of connected cubic graph on n vertices; the large orders were split into residue classes by `geng`'s `res/mod` slicing. Each pipeline then applied its own 3-connectivity filter. No graph was excluded on any other ground, and no symmetry reduction beyond `geng`'s isomorph rejection was used at any stage.

The completeness of the enumeration is the load-bearing external assumption of this note (§5), and it is supported by count agreement at every order against two independent published sources. The number of connected cubic graphs read matches the published enumeration of Brinkmann, Goedgebeur, and McKay [BGM11] — whose Table 1 gives 7,319,447 at 22 vertices and 117,940,535 at 24 — and equivalently OEIS `A002851`. The number surviving the 3-connectivity filter matches the published counts of 3-connected cubic graphs: through 20 vertices those of McKay and Royle [McKR86], and through 24 vertices OEIS `A204198`, whose terms

$$0,\ 1,\ 2,\ 4,\ 14,\ 57,\ 341,\ 2828,\ 30468,\ 396150,\ 5909292,\ 98101019,\ \ldots$$

(indexed by half the order, with $a(11)$ and $a(12)$ credited to Ed Wynn, 2023) reproduce our filtered counts exactly at every order. The two checks are independent of each other in the useful direction: the first tests the generator, the second tests the filter.

2.2. **Deciding each graph.** For a cubic graph on $n \leq 31$ vertices the whole question fits in machine words, and both pipelines decide it exactly rather than approximately. The base claim asks for a set S of $n \bmod 3$ vertices such that $G - S$ has a Λ-factor; a Λ-factor is found, or proved absent, by depth-first search over vertex triples (a, b, c) with $ab, bc \in E(G)$, extending a partial packing until the covered set is everything, with a hash cache of covered-set masks already shown to admit no completion. The strong forms are decided by running the same factor decision on

n	connected cubic	3-connected	strong forms verified	recount
4	1	1	$(f1), (f2)$	yes
6	2	2	$(z2), (z3), (z7), (z8)$	yes
8	5	4	$(t2)$	yes
10	19	14	$(f1), (f2)$	yes
12	85	57	$(z2), (z3), (z7), (z8)$	yes
14	509	341	$(t2)$	yes
16	4,060	2,828	$(f1), (f2)$	yes
18	41,301	30,468	$(z2), (z3), (z7), (z8)$	yes
20	510,489	396,150	$(t2)$	yes
22	7,319,447	5,909,292	$(f1)$	yes
≤ 22	7,875,918	6,339,157		
24	117,940,535	98,101,019	— (base claim only)	*pending*

TABLE 1. Coverage by order. The "connected cubic" column is the generator's output and matches [BGM11] and OEIS `A002851`; the "3-connected" column is what survives each pipeline's own filter and matches OEIS `A204198` at every order and [McKR86] at orders 10 through 20. Every graph counted in the third column was decided for the base claim (P) and for the strong forms listed. The last row is Proposition 1.3: complete on the search side, not yet recounted independently.

each of the linearly or quadratically many derived graphs: $G - e$ and G-with-e-forced for $(z2), (z3)$; $G - \{e, f\}$ over all $\binom{|E|}{2}$ pairs for $(z7)$; $G - V(L)$ over all 3-vertex paths for $(z8)$; $G - \{x, y\}$ over all edges for $(t2)$; $G - x$ over all vertices for $(f1)$; $G - \{x, e\}$ for $(f2)$. Because the procedure is exhaustive, a negative answer is a proof of non-existence, which is why a single failure anywhere would have been the object of interest rather than a nuisance.

The second pipeline decides the same questions by deliberately different means: connectivity by union–find rather than bitmask breadth-first search; 3-vertex-path triples precomputed into per-vertex tables rather than generated on the fly; depth-first branching on the highest uncovered vertex rather than the lowest; the base claim by explicit iteration over avoided sets rather than by a skip counter; and a four-way probed failure cache rather than a single-slot one. It shares no code with the first. Agreement between the two is therefore agreement between two search orders and two data structures, not between two runs of one program.

2.3. **Certificates.** A completed packing is a short, checkable object, and the sweeps emit it: one line per certified graph carrying the `graph6` string, the list of vertex triples forming the paths, and the avoided vertices, for example

```
CERT U???????C?W?[?Y?C`Cc?Aa?X??BG?I_?Ao?@K?? | 10-0-11 12-1-13 2-14-5
15-3-17 4-16-6 18-7-19 20-8-21 | 9
```

at order 22, where the seven triples and the single avoided vertex 9 partition the 22 vertices. Certificates were stored for every 3-connected cubic graph through order 18 and for a fixed sample at higher orders (43,580 lines through order 22; 9,776 more at order 24). A checker re-derives everything from the `graph6` string alone: it decodes the string with its own decoder, verifies that the graph is simple and cubic, optionally verifies membership of the string in a freshly generated `geng` stream for that order and re-proves 3-connectivity by exhaustive deletion of vertex pairs, checks that each listed triple is a path of the graph, and checks that the triples and the avoided vertices partition the vertex set with $|\text{avoided}| = n \bmod 3$. Certificates are not the primary evidence — full coverage rests on the searches themselves — but they make spot-checking cheap for a reader, and they are regenerable: the searchers are deterministic, so any certificate reproduces from the recorded command.

3. Independent recount, controls, and referee

3.1. **The recount.** The second pipeline re-ran the sweep from the generator forward. Through order 16 it re-solved the full stream with all strong forms enabled; at order 18 with $(z2), (z3), (z7), (z8)$; at order 20 with $(t2)$; at order 22 with $(f1)$, in eight generator slices. At every order the count of graphs read and the count surviving the 3-connectivity filter agree with the first pipeline and with the published enumerations, and the number of base-claim failures and strong-form failures is zero on every slice. At order 22 the eight slices read 518,580 + 670,736 + 982,556 + 1,068,280 + 1,569,040 + 879,959 + 978,267 + 652,029 = 7,319,447 graphs and kept 5,909,292, both exactly the published values; the order-22 recount and its certificate cross-check together consumed under two hours of single-core time on the laptop.

3.2. **Certificate cross-checks.** The certificate corpus was re-verified across the pipeline boundary. All 30,468 certificates at order 18, all 3,961 at order 20, and all 5,904 at order 22 — 40,333 in total — were accepted by the second pipeline's own checker, with zero rejections; at orders 20 and 22 the check included re-proving 3-connectivity from the `graph6` string, and at all three orders it included membership in a freshly regenerated generator stream (41,301, 510,489, and 7,319,447 lines respectively, each line count re-derived at check time). The 3,247 certificates at orders ≤ 16 were accepted by the first pipeline's independent standard-library checker, and those graphs were in addition re-solved from scratch by the second pipeline.

3.3. **Negative controls.** A sweep that only ever reports success proves nothing about its own failure path, so both tools were forced to fail. With the 3-connectivity filter disabled, the base sweep over all *connected* cubic graphs of orders 10 through 16 reports exactly one failure: the 16-vertex graph with `graph6` string `O???E?oBEAWOKGK_@o?W_`, which has a cut vertex. The same graph had been identified independently by the first pipeline's own control, and a third, unrelated brute-force implementation confirms $\lambda = 4 < 5$ for it; the two pipelines agree on the unique sub-3-connected failure in that range. With the filter disabled the strong-form paths fire as well: no failure at order 8, where none exists; 54 at order 10 (2 of type $(f1)$, 52 of $(f2)$); 253 at order 12, spread over all four $(z\cdot)$ paths (4 of $(z2)$, 8 of $(z3)$, 106 of $(z7)$, 135 of $(z8)$); 145 of $(t2)$ type at order 14; and 15,691 at order 16 (317 of $(f1)$, 15,374 of $(f2)$; the captured control log's own header line prints 15692 for this total, an arithmetic slip in the log and not in the count — its `RSUMMARY` two lines below reads `sfail=15691`). Each checker was likewise run on deliberately doctored certificates, one corruption per line: a non-path triple, a cover gap, an overlapping triple, a valid partition of sub-maximum shape, a `graph6` string with a trailing character, a `graph6` string with a flipped character, a valid certificate for a non-canonical relabelling of the graph (a string the generator never emits), and a genuine Λ-factor certificate for a connected cubic graph that is not 3-connected. Each corruption is rejected by the gate that targets it, with a nonzero exit code, while the intact certificate in the same file is accepted. The membership gate is `refcert.py`'s alone: run with `--g6set` and `--check3c` it rejects all eight, whereas `verify_cert.py`, which tests no membership in a generator stream, rejects seven and accepts the relabelling with exit code 0 (it exits on the first violation, so the control is run one line at a time). Two controls on the controls confirm the diagnosis, since disabling the membership test admits the relabelling and nothing else, and disabling the 3-connectivity test admits the sub-3-connected certificate and nothing else. All controls in this subsection were re-run on the drafting date against the current binaries and checkers.

3.4. **The adversarial referee.** The result rests on an adversarial referee pass by a separate agent instance under an explicit independence rule: every load-bearing check used code written in that session, sharing nothing with the search pipeline beyond the `graph6` format specification. The referee wrote the second pipeline, ran the recount and the cross-checks and the controls described above, recomputed every count sum mechanically from the raw per-slice summary lines rather than accepting reported totals, and fetched the published enumeration values itself. On this protocol the range $v(G) \leq 20$ was confirmed on August 6, 2026, and order 22 was recounted to completion on August 11, 2026; Theorems 1.1 and 1.2 are stated at exactly the strength those passes support. Order 24 was not recounted. What the referee did establish there is the completeness of the search side — that every job finished successfully, that the generated and filtered counts equal the published enumeration values, and that no output file anywhere contains a failure line — which is why Proposition 1.3 is phrased as a report of a completed search rather than as a verification.

4. Certification

The permanent record of the computation is the part a reader can check without us: the two certificate checkers in source form, the recorded command line for every sweep and every generator slice at every order, the per-slice summary outputs, the full certificate corpus, the negative-control inputs and outputs, and the referee's verdict records. It ships with this note in the program's public certificate repository *Certify* (`https://github.com/05oz/certify`; concept DOI 10.5281/zenodo.21799111), as Part I, version DOI 10.5281/zenodo.21897011. The two searchers are specified in §2.2 but are not part of that deposit, and we say so rather than imply otherwise: what is deposited is what re-checks their output without them, which is every positive answer they gave. Nothing large needs to be archived either — the graph streams are outputs of `geng` and regenerate from the recorded commands. To replay: checking any certificate, or the whole corpus, requires only CPython; regenerating any graph stream requires `nauty` 2.9.3. The full sweep through order 22 is about an hour and a half of single-core time per pipeline; order 24 was about 29 hours of completing-run time on the search side, and about 48 hours of machine time once the slices that exceeded their first time limit and were rerun are counted.

5. The trusted base

What a skeptic must believe, separated from what they can replay.

Machine-checked, replayable: the decision, for each of the 6,339,157 graphs, of the base claim and of the strong forms listed in Theorem 1.2, by each of two pipelines sharing no code; the count of graphs read and of graphs kept at every order, by both pipelines and against two published enumerations; all 43,580 certificates through order 22, re-verified from the `graph6` strings alone, 34,429 of them by two unrelated checkers; and the negative controls, which both pipelines' failure paths pass.

Trusted:

(1) *Completeness of the enumeration.* That `geng -c -d3 -D3` n emits at least one representative of every isomorphism class of connected cubic graph on n vertices is assumed, not proved here. It is the only assumption a counterexample could hide behind, and the mitigation is count agreement with the independent published enumerations — [BGM11] at every order and [McKR86] at orders 10 through 20, its tables stopping there — values obtained by different generators (`minibaum`, `snarkhunter`, and McKay and Royle's constructions) — and, for the filtered counts, with OEIS `A204198`. A generator that missed a class would have to miss it while still producing the published totals.

(2) *The 3-connectivity filters.* Both pipelines test 3-connectivity directly, by verifying that G, every $G - u$, and every $G - \{u, v\}$ is connected, rather than by any equivalence between vertex and edge connectivity for cubic graphs. The two implementations differ (breadth-first search on bitmasks; union–find) and agree on every count; and the standard-library checker's own 3-connectivity routine, driven over the raw generator stream by a separate

driver, reproduced the filtered counts at orders 8 through 16, closing the question in both directions at those orders.

(3) *The correctness of the exhaustive search.* A "no" from either searcher is a claim of non-existence and is not certifiable by a short witness. Nothing in Theorems 1.1 and 1.2 depends on such a "no" being right — every reported outcome is a "yes", witnessed by a packing — but the *absence of failures* does depend on the searchers not silently skipping graphs, which is why the counts, and not only the verdicts, are cross-checked at every order.

(4) *The equivalences of Theorem 1.4*, on which the counterexample-hunting reading of the strong forms rests. These are Kelmans' theorems, proved on paper in [Kel11]; they are not machine-checked here.

(5) *CPython, the C compiler, the operating system, and the hardware.*

Neither searcher is in the trusted base in the usual sense: every positive answer carries a packing that a reader can re-check without trusting the program that found it.

6. Questions

6.1. **What a structural proof would need.** The computation supplies no tightness anywhere. Every one of the 6,339,157 graphs meets the bound (1), and meets it after the deletions demanded by the strong forms as well; nothing in the range distinguishes hard instances from easy ones. That is evidence for (P) and simultaneously evidence that a proof will not come from extremal considerations at small orders.

Theorem 1.4 says a proof needs only one of the equivalent claims, and the local ones look most tractable: $(t2)$ asserts that $G - \{x, y\}$ has a Λ-factor for *every* edge xy, and $(f1)$ that $G - x$ has one for *every* vertex — universally quantified statements about a single deletion, the natural shape for a discharging or a minimal-counterexample argument. The known obstructions say where such an argument must be careful: by [Kel11, (r1)] a 3-edge cut with both sides of order not divisible by 3 kills Λ-factors of $G - X$, and by [Kel11, (r4)] claim $(t2)$ is false for non-adjacent x, y. Adjacency and the divisibility of the two sides of a cut are exactly the hypotheses that cannot be dropped.

Question 6.1. Does (P) reduce to cyclically 4- or 5-connected cubic graphs? That is, do the compositions of [Kel11, §2] let a minimum counterexample be assumed to have large cyclic connectivity, and if so, does the Λ-factor problem become tractable on that class — where, by [Kel11, 6.1], cyclically 6-connected examples with delicate behaviour already exist, from 24 vertices upward?

Question 6.2. Kelmans proves $(z1)$–$(z5)$ for cubic 3-connected *claw-free* graphs [Kel07b, Kel11b]. The class is narrow: in a cubic claw-free graph the three neighbours of any vertex cannot be pairwise non-adjacent, so every vertex lies in a triangle. What is the weakest local-structure hypothesis — a bound on the number of independent claws, say — under which the claw-free argument still runs?

6.2. **The strong forms' status.** Theorem 1.2 is complete for the forms it lists and silent elsewhere. Three gaps are immediate. Claim $(f2)$ was tested only through order 16; at order 22 its cost is the number of vertex–edge pairs times a factor decision, an order of magnitude above $(f1)$, and it was not run. Claims $(z7)$ and $(z8)$ have never been tested at order 24, which is the first order divisible by 6 beyond the verified range and therefore the first place where a $(z\cdot)$ failure would be a direct counterexample rather than a converted one. And the remaining forms of Theorem 1.4 — $(z4), (z5), (z6), (z9), (t1), (t3), (t4)$, $(f3)$–$(f6)$ — have not been tested at any order. They are equivalent to (P), so testing them adds no logical strength, but they are different searches with different failure modes, and one of them may be cheap enough to push several orders past 22 on its own. Identifying which is a concrete question.

Question 6.3. Among the claims of Theorem 1.4, which admits the cheapest exhaustive test per graph at order n? A form testable in time comparable to a single factor decision would move the frontier further than any improvement to the base search.

6.3. **Where the frontier sits.** The binding constraint is enumeration, not search. Order 24 took about 29 hours of completing-run single-core time on the search side for 117,940,535 generated graphs, a throughput near a thousand graphs per second; the outstanding work at that order is the independent recount, which is a comparable cost and would raise Proposition 1.3 to the strength of Theorem 1.1. Order 26 requires generating 2,094,480,864 connected cubic graphs and deciding 1,782,392,646 of them — roughly three weeks of single-core time per pipeline at the order-24 throughput, and more than that in truth, since per-graph cost grows with order; it is embarrassingly parallel across generator slices and so is a matter of core-days, not of algorithms. Order 26 is $\equiv 2 \bmod 6$, so it tests $(t2)$; order 28 ($\equiv 4$) tests $(f1)$ and has 35,085,504,243 graphs to decide.

The real wall is at order 30, the next order divisible by 6 after 24. OEIS `A204198` currently stops at 28 vertices, so beyond that order there is no published count of 3-connected cubic graphs to check a generator against — and count agreement with an independent enumeration is the mitigation that makes the trusted base of §5 tolerable. Past 28 vertices a sweep would be asserting its own completeness.

Question 6.4. Can the frontier be advanced much further on a restricted class? The obstructions of [Kel11, (r1)–(r4)] are cuts and short cycles, which suggests looking where neither is available; but the suggestion is not safe, since [Kel11, (r5), (r6)] exhibit obstructions to the $(f\cdot)$ forms in cubic 3-connected graphs with no 3-cycles and no 4-cycles. Generators for sparse families — `genreg` [Mer99], `snarkhunter` [BGM11] — reach far higher orders than the full cubic enumeration does, because the families are far sparser. A sweep of, say, all cyclically 5-connected cubic graphs of order $\equiv 0 \bmod 6$ up to 34 or 36 vertices looks feasible today, and would test the conjecture on the class where the compositions of [Kel11, §2] have the least room to work.

Acknowledgments

The problem is Alexander Kelmans'; the equivalence theorem, the tightness constructions, and the connection to Reed's domination conjecture that give this verification its meaning are all his, and this note is a computation performed inside his framework. The P_3-factor form of the conjecture is due to Jun-ichi Akiyama and Mikio Kano, and its continued visibility owes much to Douglas West's problem pages and to the Open Problem Garden. The enumeration rests on Brendan McKay's `nauty`, and the count cross-checks on the published cubic-graph enumerations of Brinkmann, Goedgebeur, and McKay and of McKay and Royle, and on the OEIS. The computation and drafting were AI-assisted as stated in the first footnote; the adversarial referee protocol of §3.4, its rule that a claim be stated at exactly the strength its recount supports, and its rule that a computation-only note close with the questions it opens shaped this note's final form.

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